Honors Geometry – 4.1 & 4.2 Practice

Full Name: _____

Translations and Reflections • Sections 4.1 – 4.2

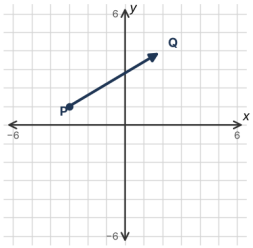
Period: _____

Show your work. Label every image with primes (A′, A″). Each part ends with a self-check, so use it before you move on.

Part A — Translations (4.1)

A translation slides EVERY point the same way. Vector ⟨a, b⟩ is the rule (x, y) → (x + a, y + b). Ex: ⟨−2, 5⟩ moves (3, 1) to (1, 6).

1. Name the vector and write its component form.



name: _____ form: ⟨___, ___⟩

2. Write a rule for the translation that moves A(2, −1) to A′(−3, 4).

(x, y) → (x + ___, y + ___)

3. Use (x, y) → (x + 4, y − 3) to find the images of J(−2, 5), K(0, 1), and L(3, −2).

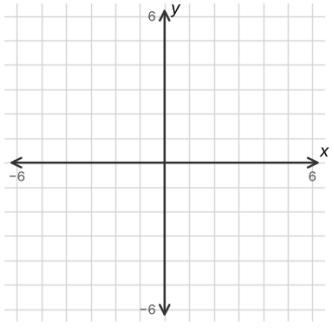
J′ _____ K′ _____ L′ _____

4. P′(6, −1) is the image of P after a translation by ⟨−2, 5⟩. Find P.

Undo it: go the opposite way.

P(___, ___)

5. Graph △ABC with A(−4, 1), B(−2, 4), C(−1, 1) and its image after a translation by ⟨5, −3⟩.



6. Translate by (x, y) → (x + 3, y − 1), then by (x, y) → (x − 5, y + 4). Write ONE rule that does both.

(x, y) → _____

✓ **Check yourself:** Add all six coordinates of your images in #3. You should get 8.

Part B — Reflections (4.2)

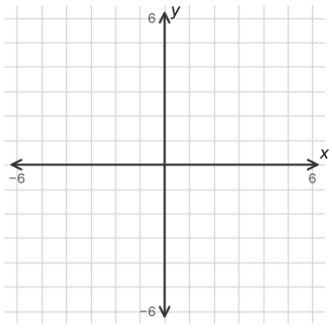
x-axis: (x, y) → (x, −y) y-axis: (x, y) → (−x, y) y = x: (x, y) → (y, x) y = −x: (x, y) → (−y, −x). For x = h or y = k: count to the line, then go the SAME distance past it.

7. Reflect the point (3, −5) in each line.

x-axis _____ y-axis _____

y = x _____ y = −x _____

8. Graph △DEF with D(1, 2), E(4, 3), F(2, 5) and its reflection in the y-axis.



9. Reflect M(−2, 3) in the line x = 1.

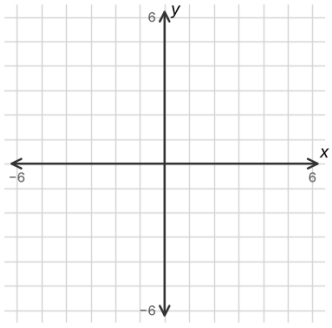
M is ___ units left of x = 1, so M′ is ___ units right.

M′(___, ___)

10. Reflect N(4, −1) in the line y = 2.

N′(___, ___)

11. Graph △QRS with Q(1, −3), R(3, −1), S(4, −4) and its reflection in the line y = x. Draw the line first.



12. A(−3, 2) maps to A′(2, −3). Name the line of reflection.

line: _____

✓ **Check yourself:** Average each point and its image in #9 and #10. You should land exactly on the line of reflection: x = 1 and y = 2.

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Part C — Glide reflections and compositions

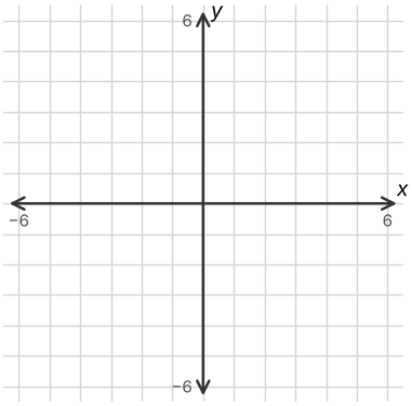
Glide reflection: a translation, THEN a reflection in a line parallel to the translation. One step at a time: write the middle image (′) before the final image (″).

13. Glide reflection: translate (x, y) → (x + 4, y), then reflect in the x-axis. Find the image of T(−3, 2).

T′ (___, ___) → T″ (___, ___)

T″(___, ___)

14. Graph △GHI with G(2, 3), H(5, 4), I(4, 1). Translate by ⟨0, −5⟩, then reflect in the y-axis. Label both images.



15. Reflect P(2, 5) in the x-axis, then translate by ⟨3, 0⟩. Now do it in the OTHER order. Do you get the same point?

_____

16. Reflect (1, 3) in the line x = 2, then reflect that image in the line x = 6. Where does it end up? What single transformation does the same thing?

point: _____ one move: _____

✓ **Check yourself:** In #15 both orders must match. If they don't, redo the reflection step, not the slide.

Part D — Push further

Same moves, hiding behind a sentence.

17. Error analysis. Marco reflected (4, −2) in the y-axis and got (4, 2). Name his mistake in one sentence. Then give the correct image.

correct image: _____

18. A translation maps △ABC to △A′B′C′. AB = 7 cm and m∠B = 40°. Find A′B′ and m∠B′. Explain why in one sentence.

A′B′ = _____ m∠B′ = _____

19. How many lines of symmetry does each figure have?

(a) regular hexagon _____

(b) rectangle that is not a square _____

(c) isosceles trapezoid _____

(d) parallelogram that is not a rectangle or rhombus _____

20. Reflect P(2, 4) in the line y = −3. Then write a rule for reflecting ANY point (x, y) in the line y = −3.

Hint: the line is halfway, so the y-values must average to −3.

P′ _____ rule: _____

✓ **Check yourself:** Your four answers in #19 should add up to 9. In #20, test your rule on P: it has to give P′.